

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination
University Theory Examination May 2018
BPH2003 Pharmaceutical Engineering

Date: 07/05/2018, Monday Time: 10: 00 a.m. to 01:00 p.m. Maximum Marks: 75

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. Which of the following is not a heat process?
[A] Drying
[B] Evaporation
[C] Crystallization
[D] Filtration
2. Mixing of following is very critical
[A] Potent drugs
[B] Abrasive materials
[C] Cohesive materials
[D] Non cohesive materials
3. Which of the following equation is not used to determine the mechanism of filtration
[A] Poiseuille
[B] Daltons
[C] Darcy
[D]Kozeny Carman
4. _____ is the ability of a metal surface to withstand repeated cycles of corrosion.
[A] Corrosion fatigue
[B] Intragranular fatigue
[C] Stress corrosion cracking
[D] Fretting corrosion

5. Which of the following is example of Alloy?
[A] Cast Iron
[B] Glass
[C] Carbon steel
[D] Stainless steel
6. Which of the following process is generally done on large scale?
[A] Trituration
[B] Levigation
[C] Pulverization by intervention
[D] Milling
7. Process used to separate insoluble particles from liquids is
[A] Drying
[B] Filtration
[C] Sieving
[D] Extraction
8. Separation of liquid by distillation is based on _____ principle
[A] Boiling Point
[B] Vapour Pressure
[C] Miscibility
[D] Viscosity
9. Fractional distillation is also known as
[A] Differential
[B] Equilibrium
[C] Flash
[D] Rectification
10. Agitator in crystallizer prevents the
[A] Formation of aggregates
[B] Formation of large crystal
[C] Formation of small crystals
[D] Spherical crystals
11. Conditions for obeying Mier's theory is
[A] Solvent and Solute are pure
[B] Solvent is volatile
[C] Temperature control is not require
[D] Close environment is not required
12. After critical moisture content _____ starts.
[A] Saturated drying region
[B] Unsaturated drying region
[C] Constant drying region
[D] Non saturated

13. Fourier law of heat conduction is best represented by
[A] $Q = -k A \, d t / d x$
[B] $Q = k A \, d x / d t$
[C] $Q = -k A$
[D] $Q = k \, d t / d x$
14. _____ is the mechanism of fluid energy mill.
[A] Inter-particulate attrition
[B] Intra-particulate attrition
[C] Glaring
[D] Glazing
15. _____ substances are mostly affected during milling due to heat generated
[A] Thermolabile
[B] Thermostable
[C] Fine powdered
[D] Fiber
16. Sublimation refers to
[A] The process of heating a solid to a high temperature to convert it to a powdery residue by drying
[B] The process of changing something into vapor
[C] The process of setting something on fire
[D] A chemical process in which a solid substance is converted directly into a gaseous substance
17. Which one of the following is the majorly involved hazard in pharmaceutical industry?
[A] Electrical
[B] Fire
[C] Mechanical
[D] Chemical
18. In cyclone separator, the separation process depends on _____
[A] Size and density
[B] Porosity
[C] Melting point
[D] Crystal type
19. Beater is an part of which mixing equipment
[A] Sigma Blade
[B] Double cone
[C] Ribbon
[D] Planetary

20. Which of blender is not considered as tumble blenders?

- [A] Twin shell blender
- [B] Double cone blender
- [C] Cubical Shaped blender
- [D] Propeller blender

SECTION – II

Q 2 Explain the filtration giving principle, factors influencing the rate of filtration and classification of equipments. Write a brief note on filter aids. **10**

Q 3 Attempt any **TWO** of the following

- A** Classify the centrifuges. Describe principle and working of a perforated basket centrifuge with labelled diagram. **05**
- B** Give the equipments for solid-solid mixing of pharmaceutical materials. Describe principle and working of a planetary mixer with labelled diagram. **05**
- C** Describe the mechanisms of size reduction with suitable examples of equipment. **05**

SECTION – III

Q 4 Attempt any **ONE** of the following.

Differentiate the following. **10**

- A**
 - 1. Batch crystallizer vs. Continuous crystallizer
 - 2. Evaporation vs. Distillation
 - 3. Orificemeter vs. Venturimeter
 - 4. Perforated centrifuge vs. Non perforated centrifuge
 - 5. Fractional distillation vs. Simple distillation
- B** Write a brief note on drying objectives, mechanism, rate of drying curve and dryers used in pharmaceutical applications. **10**

Q 5 Attempt any **FIVE** of the following.

- A** Write a brief note on Pharmaceutical hazards. **05**
- B** List the types of conveyors used for handling of solid materials in Pharma industries. Describe working of Belt conveyor with labelled diagram. **05**
- C** Which distillation apparatus is used for separation of immiscible liquids? Explain with suitable diagram. **05**
- D** Write a brief note on working, and application of multiple effect evaporator with labelled diagram. **05**
- E** Give principle, working, advantages and disadvantages of Cyclone separator with labelled diagram. **05**
- F**
 - (a) Classify the materials of pharmaceutical plant construction. **05**
 - (b) Describe briefly types of corrosion.